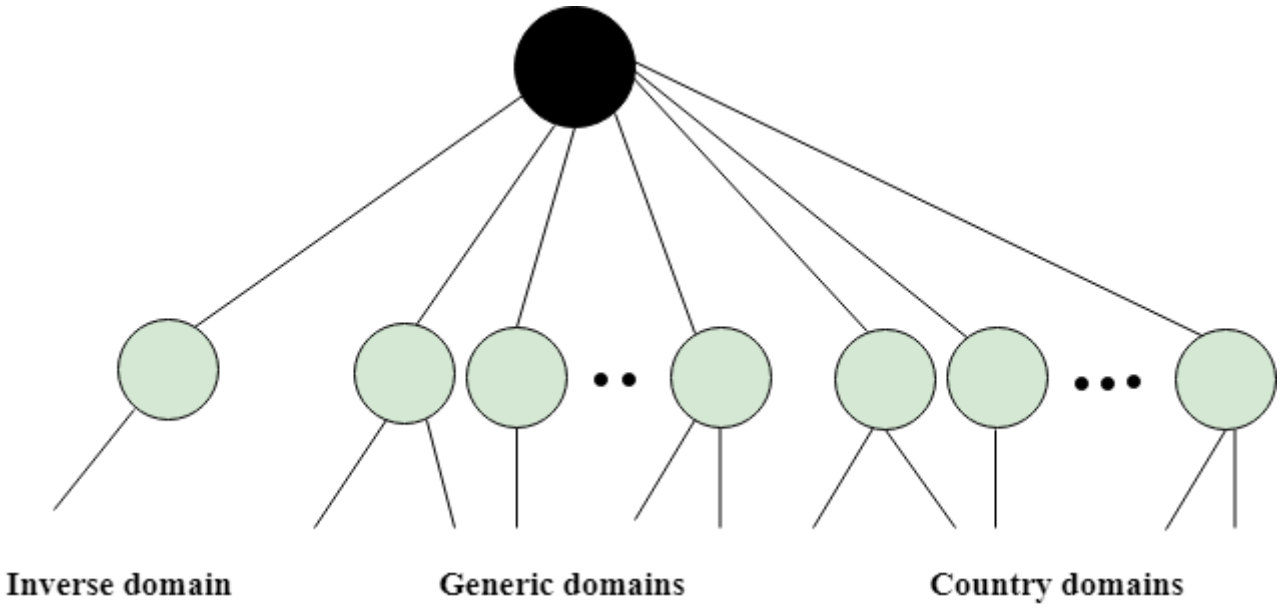
# DNS

An application layer protocol defines how the application processes running on different systems, pass the messages to each other.

- DNS stands for Domain Name System.
- DNS is a directory service that provides a mapping between the name of a host on the network and its numerical address.
- DNS is required for the functioning of the internet.
- Each node in a tree has a domain name, and a full domain name is a sequence of symbols specified by dots.
- DNS is a service that translates the domain name into IP addresses. This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses.
- For example, suppose the FTP site at EduSoft had an IP address of 132.147.165.50, most people would reach this site by specifying ftp.EduSoft.com. Therefore, the domain name is more reliable than IP address.

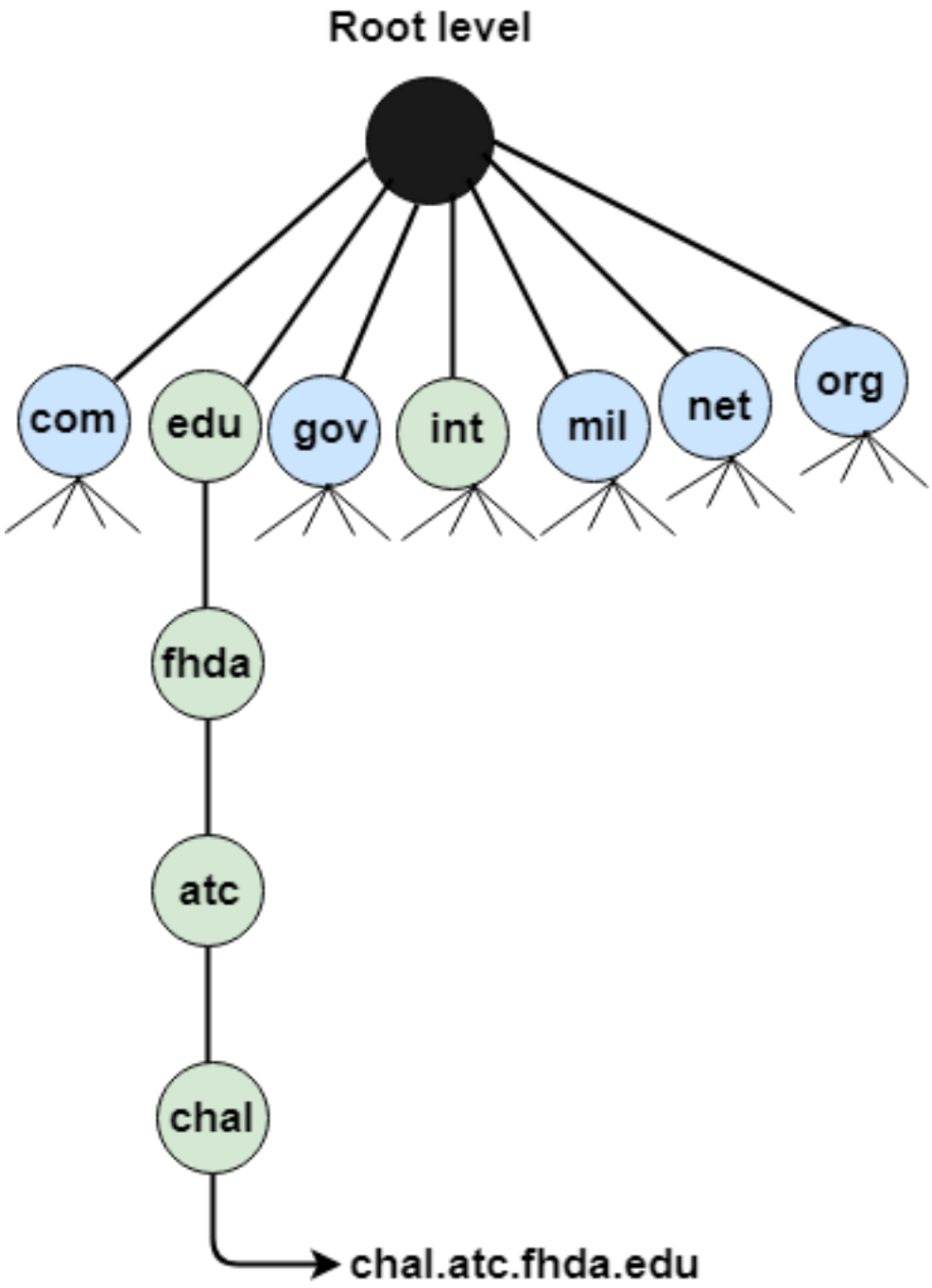
DNS is a TCP/IP protocol used on different platforms. The domain name space is divided into three different sections: generic domains, country domains, and inverse domain.



# Generic Domains

- It defines the registered hosts according to their generic behavior.
- Each node in a tree defines the domain name, which is an index to the DNS database.
- It uses three-character labels, and these labels describe the organization type.

| Label | Description |
|---|---|
| aero | Airlines and aerospace companies |
| biz | Businesses or firms |
| com | Commercial Organizations |
| coop | Cooperative business Organizations |
| edu | Educational institutions |
| gov | Government institutions |
| info | Information service providers |
| int | International Organizations |
| mil | Military groups |
| museum | Museum & other nonprofit organizations |
| name | Personal names |
| net | Network Support centers |
| org | Nonprofit Organizations |
| pro | Professional individual Organizations |




## Country Domain

The format of country domain is same as a generic domain, but it uses two-character country abbreviations (e.g., us for the United States) in place of three character organizational abbreviations.

## Inverse Domain

The inverse domain is used for mapping an address to a name. When the server has received a request from the client, and the server contains the files of only authorized clients. To determine whether the client is on the authorized list or not, it sends a query to the DNS server and ask for mapping an address to the name.

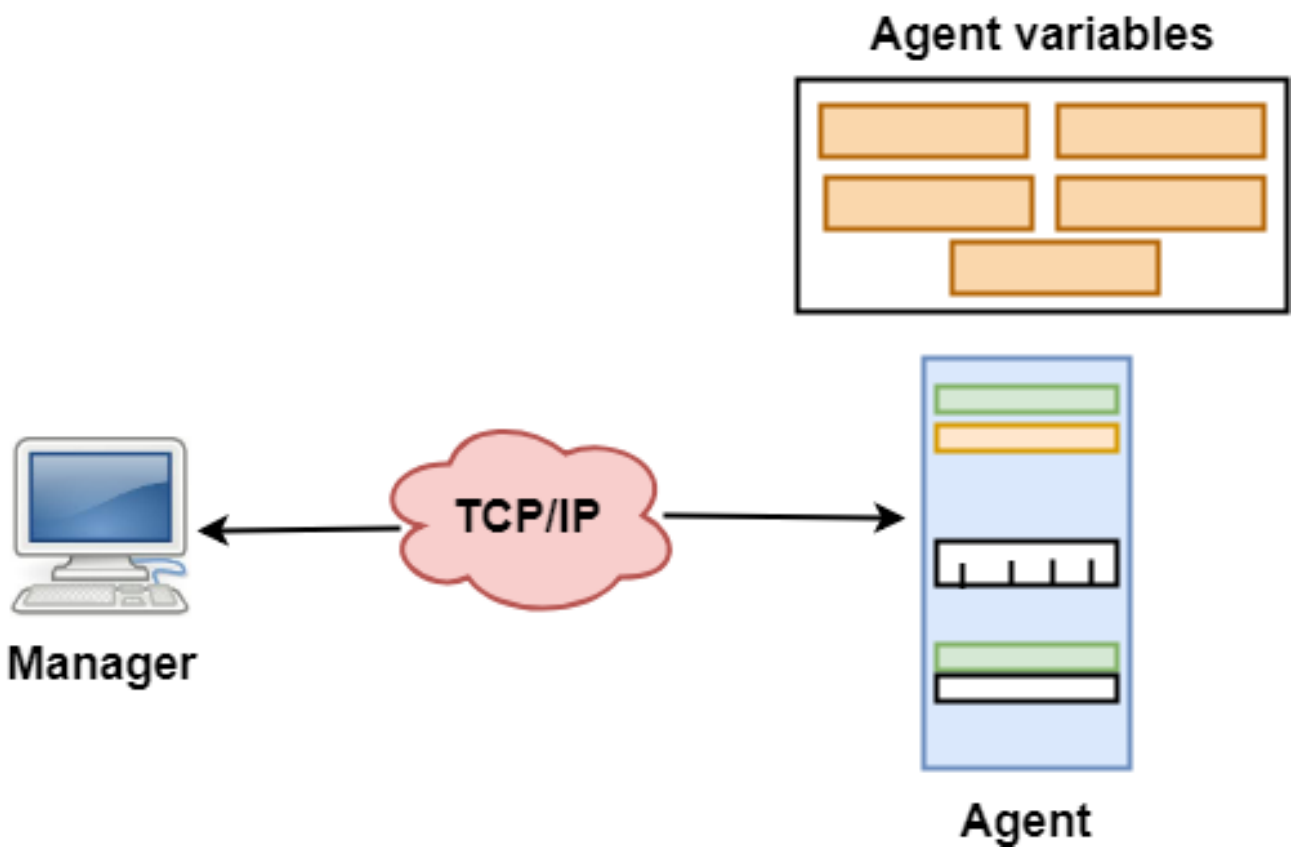
## Working of DNS

- DNS is a client/server network communication protocol. DNS clients send requests to the. server while DNS servers send responses to the client.
- Client requests contain a name which is converted into an IP address known as a forward DNS lookups while requests containing an IP address which is converted into a name known as reverse DNS lookups.
- DNS implements a distributed database to store the name of all the hosts available on the internet.
- If a client like a web browser sends a request containing a hostname, then a piece of software such as **DNS resolver** sends a request to the DNS server to obtain the IP address of a hostname. If DNS server does not contain the IP address associated with a hostname, then it forwards the request to another DNS server. If IP address has arrived at the resolver, which in turn completes the request over the internet protocol.

# SNMP

- SNMP stands for **Simple Network Management Protocol**.
- SNMP is a framework used for managing devices on the internet.
- It provides a set of operations for monitoring and managing the internet.

## SNMP Concept




- SNMP has two components Manager and agent.
- The manager is a host that controls and monitors a set of agents such as routers.
- It is an application layer protocol in which a few manager stations can handle a set of agents.
- The protocol designed at the application level can monitor the devices made by different manufacturers and installed on different physical networks.
- It is used in a heterogeneous network made of different LANs and WANs connected by routers or gateways.

# Managers & Agents

- A manager is a host that runs the SNMP client program while the agent is a router that runs the SNMP server program.
- Management of the internet is achieved through simple interaction between a manager and agent.
- The agent is used to keep the information in a database while the manager is used to access the values in the database. For example, a router can store the appropriate variables such as a number of packets received and forwarded while the manager can compare these variables to determine whether the router is congested or not.

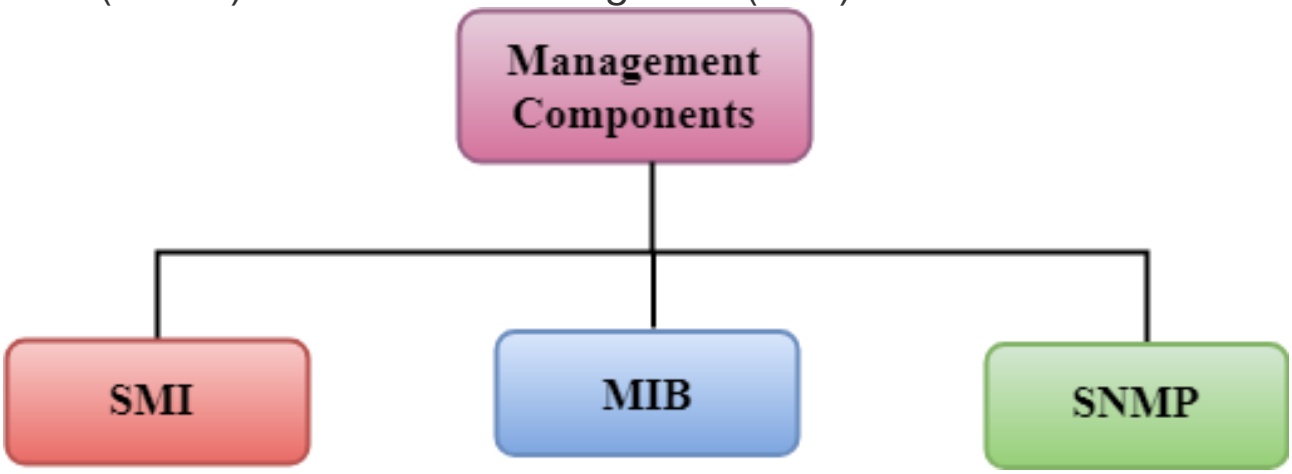
- Agents can also contribute to the management process. A server program on the agent checks the environment, if something goes wrong, the agent sends a warning message to the manager.

# Management with SNMP has three basic ideas:

- A manager checks the agent by requesting the information that reflects the behavior of the agent.
- A manager also forces the agent to perform a certain function by resetting values in the agent database.
- An agent also contributes to the management process by warning the manager regarding an unusual condition.

# Management Components

- Management is not achieved only through the SNMP protocol but also the use of other protocols that can cooperate with the SNMP protocol. Management is achieved through the use of the other two protocols: SMI (Structure of management information) and MIB(management information base).
- Management is a combination of SMI, MIB, and SNMP. All these three protocols such as abstract syntax notation 1 (ASN.1) and basic encoding rules (BER).

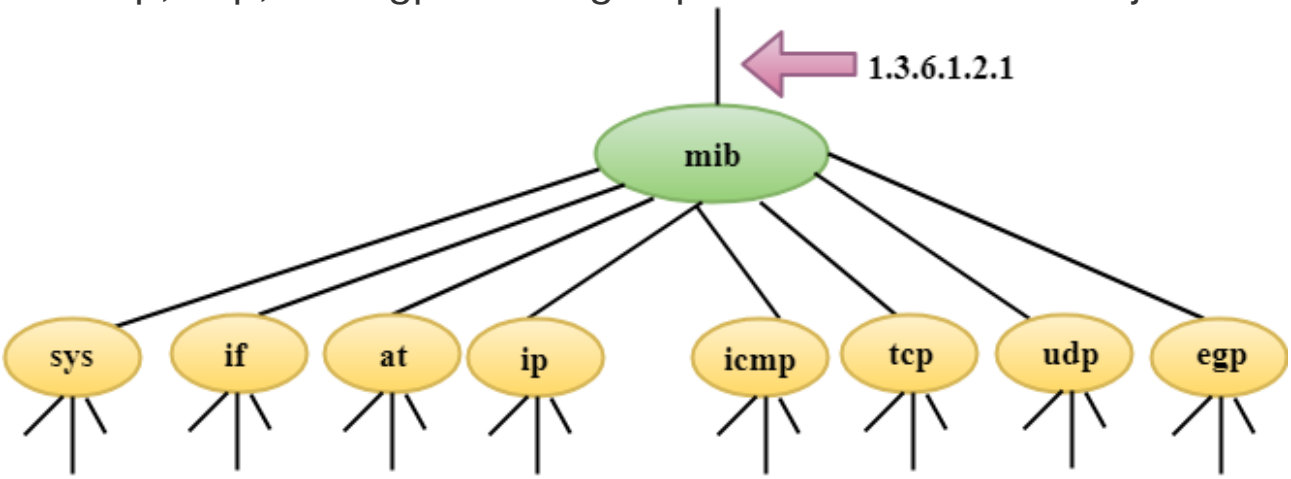



## SMI

The SMI (Structure of management information) is a component used in network management. Its main function is to define the type of data that can be stored in an object and to show how to encode the data for the transmission over a network.

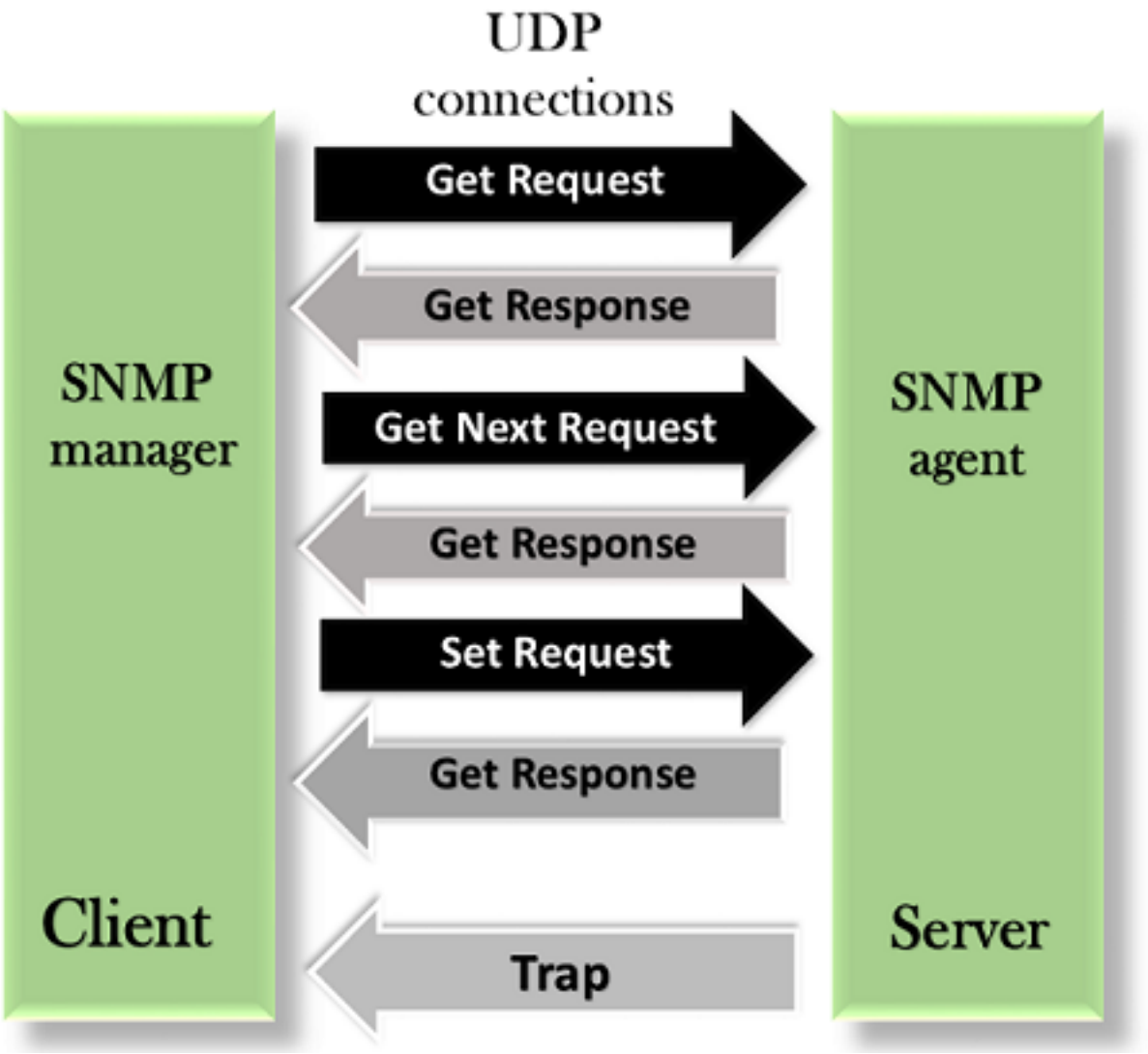
## MIB

- The MIB (Management information base) is a second component for the network management.
- Each agent has its own MIB, which is a collection of all the objects that the manager can manage. MIB is categorized into eight groups: system, interface, address translation, ip, icmp, tcp, udp, and egp. These groups are under the mib object.




## SNMP

SNMP defines five types of messages: GetRequest, GetNextRequest, SetRequest, GetResponse, and Trap.




**GetRequest:** The GetRequest message is sent from a manager (client) to the agent (server) to retrieve the value of a variable.

**GetNextRequest:** The GetNextRequest message is sent from the manager to agent to retrieve the value of a variable. This type of message is used to retrieve the values of the entries in a table. If the manager does not know the indexes of the entries, then it will not be able to retrieve the values. In such situations, GetNextRequest message is used to define an object.

**GetResponse:** The GetResponse message is sent from an agent to the manager in response to the GetRequest and GetNextRequest message. This message contains the value of a variable requested by the manager.

**SetRequest:** The SetRequest message is sent from a manager to the agent to set a value in a variable.

**Trap:** The Trap message is sent from an agent to the manager to report an event. For example, if the agent is rebooted, then it informs the manager as well as sends the time of rebooting.

## Electronic mail, commonly known as email, is a method of exchanging messages over the internet. Here are the basics of email:

1. An email address: This is a unique identifier for each user, typically in the format of name@domain.com.
2. An email client: This is a software program used to send, receive and manage emails, such as Gmail, Outlook, or Apple Mail.
3. An email server: This is a computer system responsible for storing and forwarding emails to their intended recipients.

## To send an email:

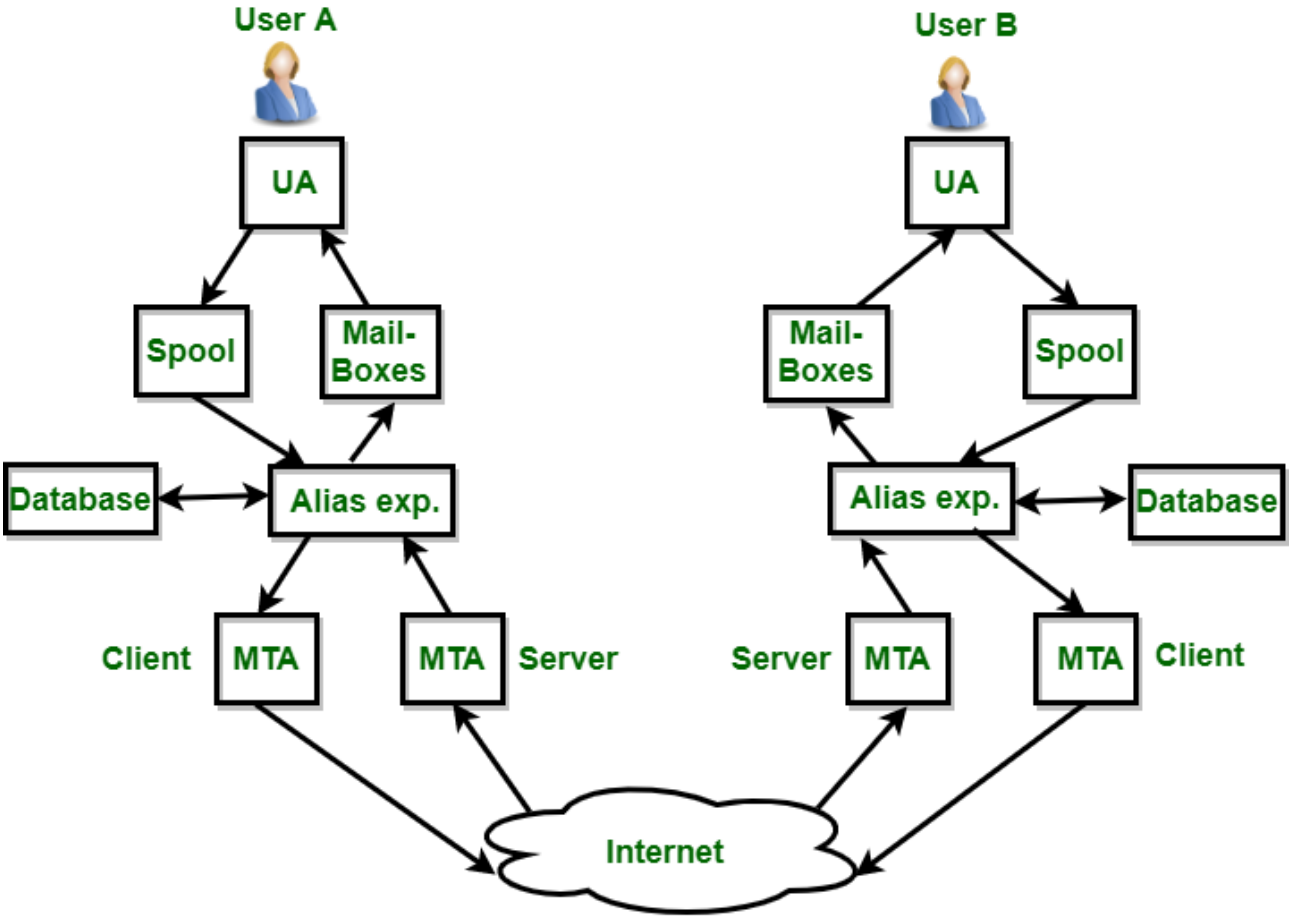
1. Compose a new message in your email client.
2. Enter the recipient's email address in the "To" field.
3. Add a subject line to summarize the content of the message.
4. Write the body of the message.
5. Attach any relevant files if needed.
6. Click "Send" to deliver the message to the recipient's email server.
7. Emails can also include features such as cc (carbon copy) and bcc (blind carbon copy) to send copies of the message to multiple recipients, and reply, reply all, and forward options to manage the conversation.

**Electronic Mail** (e-mail) is one of most widely used services of Internet. This service allows an Internet user to send a **message in formatted manner (mail)** to the other Internet user in any part of world. Message in mail not only contain text, but it also contains images, audio and videos data. The person

who is sending mail is called **sender** and person who receives mail is called **recipient**. It is just like postal mail service.
**Components of E-Mail System :** The basic components of an email system are : User Agent (UA), Message Transfer Agent (MTA), Mail Box, and Spool file. These are explained as following below.

1. **User Agent (UA) :** The UA is normally a program which is used to send and receive mail. Sometimes, it is called as mail reader. It accepts variety of commands for composing, receiving and replying to messages as well as for manipulation of the mailboxes.
2. **Message Transfer Agent (MTA) :** MTA is actually responsible for transfer of mail from one system to another. To send a mail, a system must have client MTA and system MTA. It transfer mail to mailboxes of recipients if they are connected in the same machine. It delivers mail to peer MTA if destination mailbox is in another machine. The delivery from one MTA to another MTA is done by Simple

Mail Transfer Protocol.




3. **Mailbox :** It is a file on local hard drive to collect mails. Delivered mails are present in this file. The user can read it delete it according to his/her requirement. To use e-mail system each user must have a mailbox . Access to mailbox is only to owner of mailbox.
4. **Spool file :** This file contains mails that are to be sent. User agent appends outgoing mails in this file using SMTP. MTA extracts pending mail from spool file for their delivery. E-mail allows one name, an **alias**, to represent several different e-mail addresses. It is known as **mailing list**, Whenever user have to sent a message, system checks recipient's name against alias database. If mailing list is present for defined alias, separate messages, one for each entry in the list, must be prepared and handed to MTA. If for defined alias, there is no such mailing list is present, name itself becomes naming address and a single message is delivered to mail transfer entity.

**Services provided by E-mail system :**

- **Composition –** The composition refer to process that creates messages and answers. For composition any kind of text editor can be used.
- **Transfer –** Transfer means sending procedure of mail i.e. from the sender to recipient.
- **Reporting –** Reporting refers to confirmation for delivery of mail. It help user to check whether their mail is delivered, lost or rejected.
- **Displaying –** It refers to present mail in form that is understand by the user.
- **Disposition –** This step concern with recipient that what will recipient do after receiving mail i.e save mail, delete before reading or delete after reading.

## Advantages Or Disadvantages:

## Advantages of email:

1. Convenient and fast communication with individuals or groups globally.
2. Easy to store and search for past messages.
3. Ability to send and receive attachments such as documents, images, and videos.
4. Cost-effective compared to traditional mail and fax.
5. Available 24/7.

## Disadvantages of email:

1. Risk of spam and phishing attacks.
2. Overwhelming amount of emails can lead to information overload.
3. Can lead to decreased face-to-face communication and loss of personal touch.
4. Potential for miscommunication due to lack of tone and body language in written messages.
5. Technical issues, such as server outages, can disrupt email service.

6. It is important to use email responsibly and effectively, for example, by keeping the subject line clear and concise, using proper etiquette, and protecting against security threats.

# World Wide Web (WWW)

The **World Wide Web** is abbreviated as WWW and is commonly known as the web. The WWW was initiated by CERN (European library for Nuclear Research) in 1989.

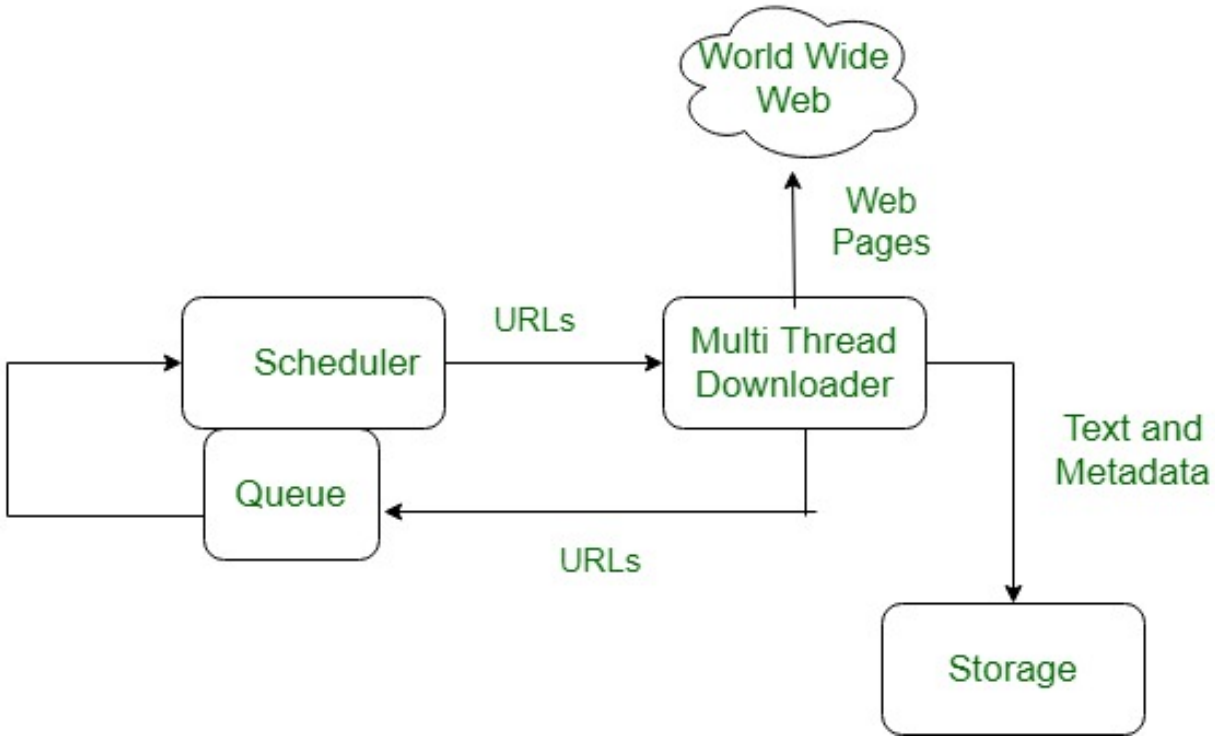
WWW can be defined as the collection of different websites around the world, containing different information shared via local servers(or computers).

**History:**
It is a project created, by Timothy Berner Lee in 1989, for researchers to work together effectively at CERN. is an organization, named the World Wide Web Consortium (W3C), which was developed for further development of the web. This organization is directed by Tim Berner's Lee, aka the father of the web.

**System Architecture:**
From the user's point of view, the web consists of a vast, worldwide connection of documents or web pages. Each page may contain links to other pages anywhere in the world. The pages can be retrieved and viewed by using browsers of which internet explorer, Netscape Navigator, Google Chrome, etc are the popular ones. The browser fetches the page requested interprets the text and formatting commands on it, and displays the page, properly formatted, on the screen.

The basic model of how the web works are shown in the figure below. Here the browser is displaying a web page on the client machine. When the user clicks on a line of text that is linked to

a page on the abd.com server, the browser follows the hyperlink by sending a message to the abd.com server asking it for the page.




Here the browser displays a web page on the client machine when the user clicks on a line of text that is linked to a page on abd.com, the browser follows the hyperlink by sending a message to the abd.com server asking for the page.

**Working of WWW:**
The World Wide Web is based on several different technologies: Web browsers, Hypertext Markup Language (HTML) and Hypertext Transfer Protocol (HTTP).

A Web browser is used to access web pages. Web browsers can be defined as programs which display text, data, pictures, animation and video on the Internet. Hyperlinked resources on the World Wide Web can be accessed using software interfaces provided by Web browsers. Initially, Web browsers were used only for surfing the Web but now they have become more universal. Web browsers can be used for several tasks

including conducting searches, mailing, transferring files, and much more. Some of the commonly used browsers are Internet Explorer, Opera Mini, and Google Chrome.

**Features of WWW:**

- HyperText Information System
- Cross-Platform
- Distributed
- Open Standards and Open Source
- Uses Web Browsers to provide a single interface for many services
- Dynamic, Interactive and Evolving.
- “Web 2.0”

**Components of the Web:** There are 3 components of the web:

1. **Uniform Resource Locator (URL):** serves as a system for resources on the web.
2. **HyperText Transfer Protocol (HTTP):** specifies communication of browser and server.
3. **Hyper Text Markup Language (HTML):** defines the structure, organisation and content of a webpage.

# HTTP Full Form

**HTTP** stands for HyperText Transfer Protocol. Tim Berner invents it. HyperText is the type of text which is specially coded with the help of some standard coding language called HyperText Markup Language (HTML). **HTTP/2** is the successor version of HTTP, which was published on May 2015. HTTP/3 is the latest version of HTTP, which is published in 2022.

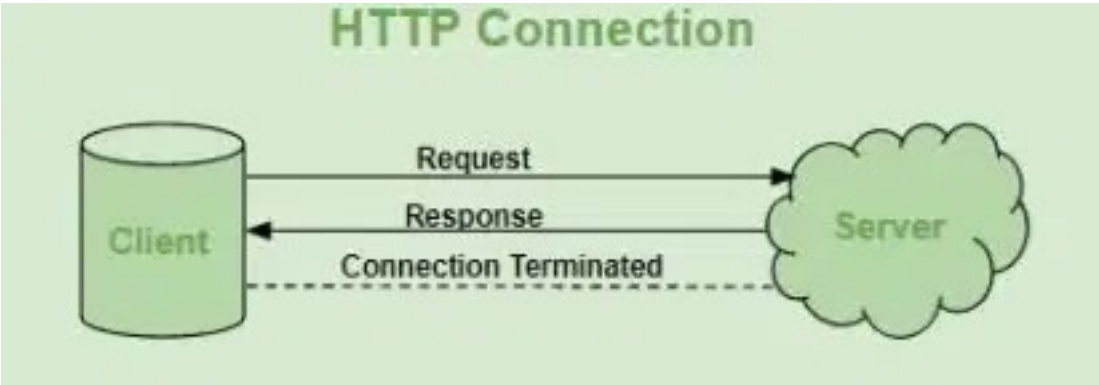
The protocol used to transfer hypertext between two computers is known as HyperText Transfer Protocol.
HTTP provides a standard between a web browser and a web

server to establish communication. It is a set of rules for transferring data from one computer to another. Data such as text, images, and other multimedia files are shared on the World Wide Web. Whenever a web user opens their web browser, the user indirectly uses HTTP. It is an application protocol that is used for distributed, collaborative, hypermedia information systems.

## Working of HTTP

First of all, whenever we want to open any website then first open a web browser after that we will type the URL of that website (e.g., www.facebook.com ). This URL is now sent to Domain Name Server (DNS). Then DNS first check records for this URL in their database, then DNS will return the IP address to the web browser corresponding to this URL. Now the browser is able to send requests to the actual server.

After the server sends data to the client, the connection will be closed. If we want something else from the server we should have to re-establish the connection between the client and the server.




HTTP Connection

## HTTP Request

HTTP request is simply termed as the information or data that is needed by Internet browsers for loading a website. This is simply known as HTTP Request.

There is some common information that is generally present in all HTTP requests. These are mentioned below.

- HTTP Version
- URL
- HTTP Method
- HTTP Request Headers
- HTTP Body

### HTTP Request Headers

HTTP Request Headers generally store information in the form of key-value and must be present in each HTTP Request. The use of this Request Header is to provide core information about the client's information, etc.

### HTTP Request Body

HTTP Request Body simply contains the information that has to be transferred. HTTP Request has the information or data to be sent to these browsers.

### HTTP Method

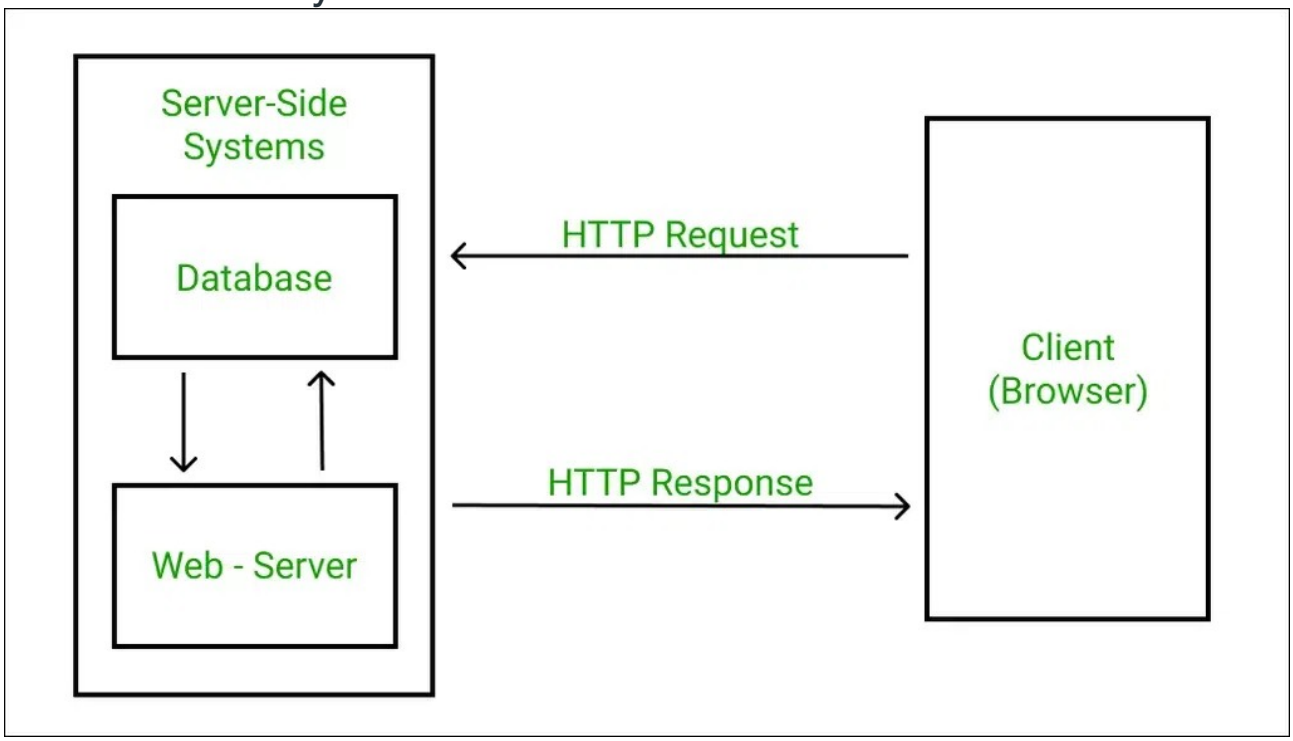
HTTP Methods are simply HTTP Verb. In spite of being present so many HTTP Methods, the most common HTTP Methods are HTTP GET and HTTP POST. These two are generally used in HTTP cases. In HTTP GET, the information is received in the form of a website.

For more, refer to the Difference Between HTTP GET and HTTP POST.

## HTTP Response

HTTP Response is simply the answer to what a Server gets when the request is raised. There are various things contained in HTTP Response, some of them are listed below.

- HTTP Status Code
- HTTP Headers
- HTTP Body




HTTP Response

### HTTP Response Headers

HTTP Response headers are simply like an HTTP Request where it has that work to send some important files and data to the HTTP Response Body.

### HTTP Response Body

HTTP Responses are the responses that are received successfully upon the request. Generally, it comes under the requests generated by the web. In most cases, the request is of transferring the HTML data into a webpage.

## HTTP Status Code

HTTP Status Codes are the 3-Digit codes that tell the message or simply tell us about the HTTP Request whether it has been completed or not. There are simply 5 types of status codes.

- Informational
- Successful
- Re-directional
- Client-Error
- Server-Error

For more, refer to HTTP Status Codes.

# History of HTTP

Tim Berners Lee and his team at CERN get credit for inventing original HTTP and associated technologies.

- **HTTP version 0.9:** This was the first version of HTTP which was introduced in 1991.
- **HTTP version 1.0:** In 1996, RFC 1945 (Request For Comments) was introduced in HTTP version 1.0.
- **HTTP version 1.1:** In January 1997, RFC 2068 was introduced in HTTP version 1.1. Improvements and updates to the HTTP version 1.1 standard were released under RFC 2616 in June 1999.
- **HTTP version 2.0:** The HTTP version 2.0 specification was published as RFC 7540 on May 14, 2015.
- **HTTP version 3.0:** HTTP version 3.0 is based on the previous RFC draft. It is renamed as Hyper-Text Transfer Protocol QUIC which is a transport layer network protocol developed by Google.

# Characteristics of HTTP

HTTP is IP based communication protocol that is used to deliver data from server to client or vice-versa.

- The server processes a request, which is raised by the client, and also server and client know each other only during the current bid and response period.
- Any type of content can be exchanged as long as the server and client are compatible with it.
- Once data is exchanged, servers and clients are no longer connected.

- It is a request and response protocol based on client and server requirements.
- It is a connection-less protocol because after the connection is closed, the server does not remember anything about the client and the client does not remember anything about the server.
- It is a stateless protocol because both client and server do not expect anything from each other but they are still able to communicate.

## Advantages of HTTP

- Memory usage and CPU usage are low because of fewer simultaneous connections.
- Since there are few TCP connections hence network congestion is less.
- Since handshaking is done at the initial connection stage, then latency is reduced because there is no further need for handshaking for subsequent requests.
- The error can be reported without closing the connection.
- HTTP allows HTTP pipe-lining of requests or responses.

## Disadvantages of HTTP

- HTTP requires high power to establish communication and transfer data.
- HTTP is less secure because it does not use any encryption method like HTTPS and use TLS to encrypt regular HTTP requests and response.
- HTTP is not optimized for cellular phones and it is too gabby.
- HTTP does not offer a genuine exchange of data because it is less secure.
- The client does not close the connection until it receives complete data from the server; hence, the server needs to wait for data completion and cannot be available for other clients during this time.

# STREAMING AUDIO AND VIDEO:

Streaming audio and video refer to the delivery of multimedia content, such as music and videos, in a continuous and real-time manner over the internet. Unlike traditional methods where you need to download an entire file before playing it, streaming allows you to start playback almost instantly without waiting for the entire file to be downloaded.

Here's an overview of streaming audio and video:

1. **Streaming Protocols:**
    - **HTTP Streaming (HLS and DASH):** HTTP Live Streaming (HLS) and Dynamic Adaptive Streaming over HTTP (DASH) are two popular protocols for streaming. They break the content into small, manageable chunks and deliver them over regular HTTP connections. This allows adaptive streaming, where the quality of the content can be adjusted based on the viewer's internet speed and device capabilities.

    - **RTMP (Real-Time Messaging Protocol):** Previously popular for live streaming, RTMP has been widely used for delivering audio and video content. However, it is being phased out in favor of HTTP-based protocols.

    - **RTSP (Real-Time Streaming Protocol):** RTSP is a network control protocol used for establishing and controlling the media stream between the server and the client. It's commonly used for streaming video.

    - **WebRTC (Web Real-Time Communication):** WebRTC is a free, open-source project that enables real-time communication of audio, video, and data directly between browsers. It's commonly used for video conferencing and live streaming applications.

2. **Streaming Services:**

- **Audio Streaming Services:** Platforms like Spotify, Apple Music, and Amazon Music use streaming to deliver music to users. Users can listen to tracks without downloading them and create personalized playlists.

- **Video Streaming Services:** Services like Netflix, YouTube, Hulu, and Amazon Prime Video stream movies, TV shows, and other video content. Users can watch content on-demand without waiting for the entire file to download.

3. **Adaptive Streaming:**
   - **Adaptive Bitrate Streaming (ABR):** ABR adjusts the quality of the streamed content in real-time based on the viewer's network conditions. If the internet connection is slow, the stream adapts by lowering the quality to prevent buffering.

4. **Live Streaming:**
   - **Live Streaming Platforms:** Services like Twitch, YouTube Live, and Facebook Live enable users to broadcast live video content. This is widely used for events, gaming, tutorials, and other real-time experiences.

5. **Challenges:**
   - **Latency:** Achieving low latency is crucial, especially for live streaming applications. The delay between the actual event and its display on the viewer's screen should be minimized.

   - **Buffering:** Buffering occurs when the playback device doesn't receive data quickly enough to continue playing without interruption. Adaptive streaming helps mitigate buffering by adjusting the quality based on the available bandwidth.

Streaming has revolutionized the way we consume multimedia content, providing convenient, on-demand access to a vast array of audio and video material over the internet.